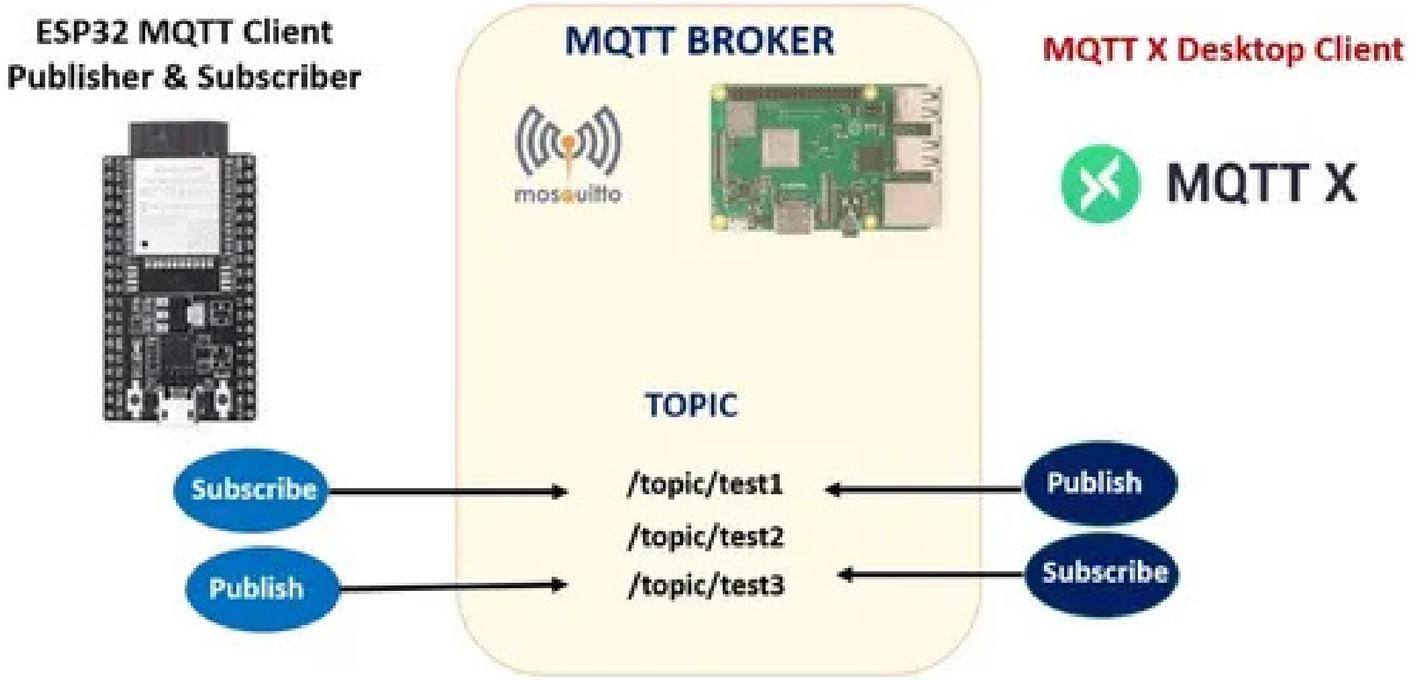
MQTT Protocol

Lightweight, Publish-Subscribe Messaging for ESP32 IoT Communication

What Is MQTT?

A lightweight messaging protocol that allows ESP32 devices and applications to exchange data through a central MQTT broker.



Publish / Subscribe

Devices publish messages to topics and subscribers receive messages from those topics.

Central MQTT Broker

The broker receives published messages and delivers them to clients that are subscribed to the corresponding topic.

Topic-Based Communication

Messages are organized using topics such as: esp32/trial/out and esp32/trial/in

Lightweight IoT Messaging

MQTT is designed for efficient messaging between connected devices and applications.

Where Is MQTT Used?

Anywhere connected devices need to exchange small messages through a reliable messaging infrastructure.

IoT Devices · **Sensor Monitoring** · **Smart Homes** · **Remote Control** · **Device Telemetry** · **Web Dashboards**

Prerequisites & Hardware :

Hardware: One ESP32 Development Board (e.g., DOIT ESP32 DEVKIT V1)

USB data cable: Required for programming and Serial Monitor access. Make sure the cable supports data, not only charging.

Power: USB power is sufficient for normal testing. A separate regulated supply can be used if the board's power-input requirements are satisfied.

Software: Arduino IDE installed on your computer

Network Requirements: A 2.4 GHz Wi-Fi Network (SSID and Password)

Step 1: Configure the Arduino IDE

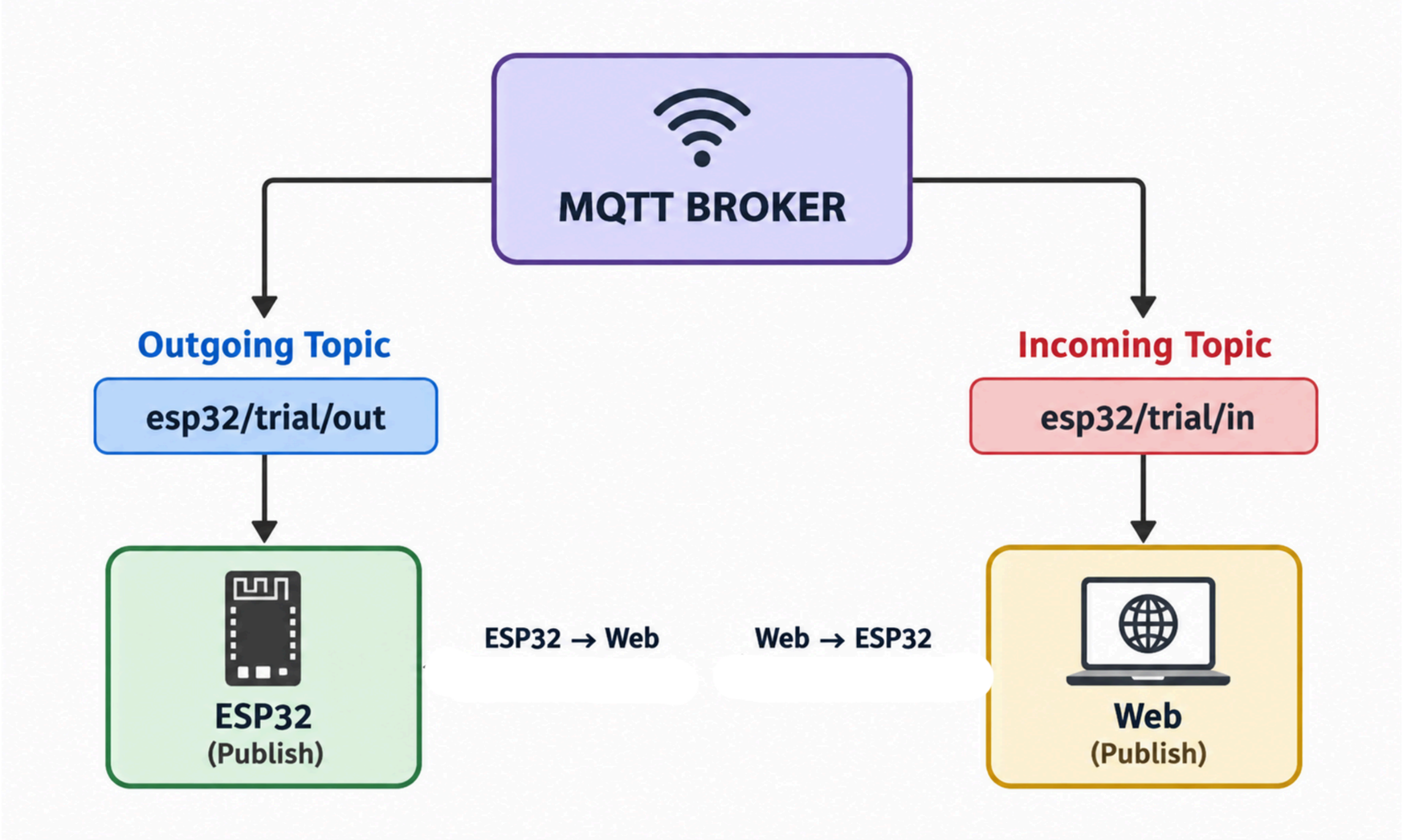
Step 2: Upload the ESP32 MQTT Code

Connect to the MQTT Broker

Step 4: Open the Web Test Dashboard

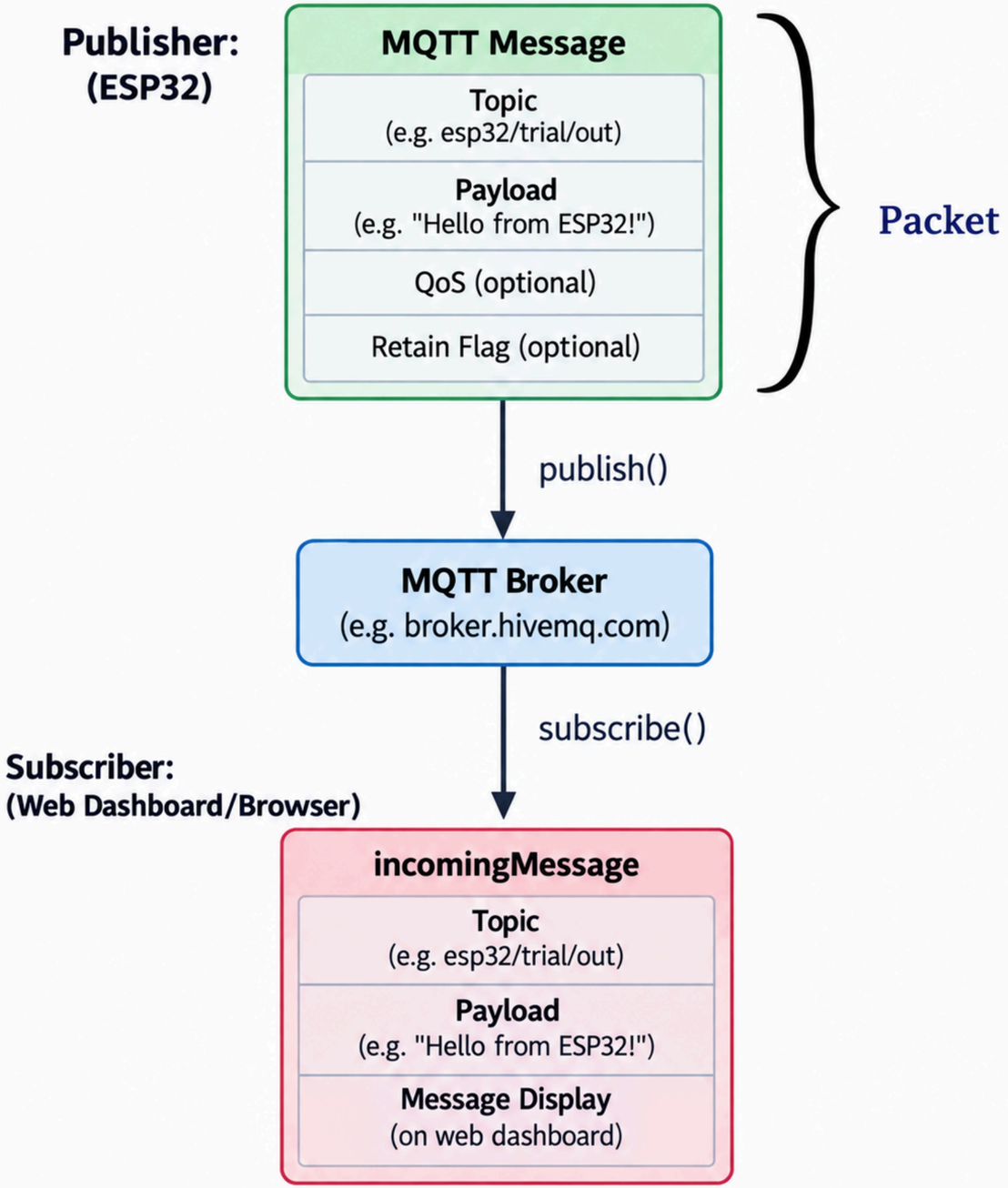
Step 5: Run the Trial and Verify

Understand MQTT Topics



A topic acts as the named channel through which MQTT messages are published and received.

Understand the MQTT Message Flow



That's MQTT Communication!

The access the implementation guide
Scan the QR / click on the link in the description

Thank You